KHULNA UNIVERSITY OF ENGINEERING & TECHNOLOGY
B.Sc. Engineering 4th Year 1st Term Examination, 2019
Department of Computer Science and Engineering
CSE 4127
Image Processing and Computer Vision

TIME: 3 hours FULL MARKS: 210

N.B. i) Answer **ANY THREE** questions from each section in separate scripts.
ii) Figures in the right margin indicate full marks.

SECTION A

(Answer **ANY THREE** questions from this section in Script A)

1. a) Specify image compression. 'Huffman Coding can be used as a symbol encoder in an image compression system'-illustrate the statement. (12)
   b) Explain false contouring with necessary examples using the concept of sampling and quantization. (08)
   c) Consider the image segment shown below: (10)

4	2	3	2 (q)
3	3	1	3
2	3	2	2
(p) 2	1	2	3

Complete length of the shortest-4, shortest-8, and shortest-m paths between pixels p & q where, v = {1, 2}. If a particular path does not exist between these two points, explain why?
   d) Interpret the significance of shading correction in digital image processing. (05)

2. a) Construct a modified image using histogram specification where histogram of original and target images are given below: (20)
original histogram:

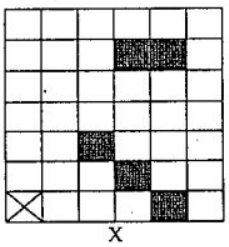
Gray level	0	1	2	3	4	5	6	7
No. of Pixels	8	10	10	2	12	16	4	2

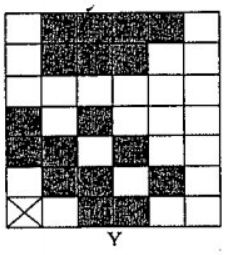
Target histogram:

Gray level	0	1	2	3	4	5	6	7
No. of Pixels	0	0	0	0	20	20	16	8

   b) Assuming continuous intensity values, suppose that an image has the intensity PDF $P_r(r) = {}^{2r}/_{(L-1)^2}$ for $0 \le r \le (L-1)$ and $P_r(r) = 0$ for other values of r. find the transformation function that will produce an image whose intensity PDF is $P_z(z) = {}^{3z^2}/_{(L-1)^3}$ for $0 \le z \le (L-1)$ and $P_z(z) = 0$ for other values of z. (10)
   c) 'Local Histogram processing is necessary to enhance details over small areas in an image'-justify the statement. (05)

3. a) Image X and Y are shown below with crossing in the left bottom showing their correspondence. (12)


X


Y

Design an appropriate structure element such that Y can be obtained after dilating X. (Black for 1 and white for 0).
   b) 'Textural segmentation is crucial in finding a boundary between two regions based on textural content'-justify the statement with necessary figures. (11)
   c) Detect edge on the following image using erosion and show the outcome of each step separately. (12)

0	0	1	1	1	1	0	1	1	1	0
0	1	1	1	1	1	1	1	1	1	0
0	1	1	1	1	1	1	1	1	1	0
1	1	1	1	0	1	1	1	1	1	1
0	1	1	1	1	1	1	1	1	1	1
0	1	1	1	1	1	1	1	1	1	0
0	1	1	1	1	1	1	1	1	1	0
0	0	0	0	0	1	1	1	0	0	0

0	1	0
1	1	1
0	1	0

Structuring Element

4. a) State and explain the data redundancies that can be identified in an image for the exploitation of image compression. (09)
   b) Define Hue and Saturation. Calculate the HSI value of the following RGB point: (11)
   (155, 210, 75)
   c) What will happen if you change the value of H channel of an HSI image while keeping the other channel's value constant? (05)
   d) Illustrate the concepts: (i) Color slicing and (ii)Tone correction (10)

SECTION B

(Answer **ANY THREE** questions from this section in Script B)

5. a) Compare between image processing and computer vision. Explain some typical challenges in solving a computer vision problem. (12)
   b) Consider the 1D image below: (08)

a	b	c	d	e	f	g

   i) Suppose, you apply a width-3 mean filter twice. Give an expression for the center pixel after the filter has been applied.
   ii) Define a single 1D kernel that, when applied only once to the image, will produce the same results.
   c) Find the unsharp mask first and then the sharped image for the following image: (15)

2	8	6
5	4	1
7	3	9

   Use a 3x5 mean filter as a blurring kernel and do wrap around padding where required.

6. a) Design a Bulterworth Low-pass Filter (BLPF). What happens when you choose a higher order BLPF to smooth an image? When would you prefer BLPF than Gaussian Low-pass Filter? (12)
   b) Show that the Fourier transform of a shifted function is unaltered except for a linearly varying phase factor. (08)
   c) Verify the soundness of the statement: 'Laplacian with a -8 in the center yields sharper results than the one with -4 in the center'. (10)
   d) Explain estimation of image degradation function by mathematical modeling. (05)

7. a) Segment the image shown below with a global threshold computed by Otsu's method. (13)

1	2	0	12	14
2	3	4	8	13
9	2	3	15	7
3	2	2	11	13
1	2	0	14	12

   Mention some limitations of global thresholding.
   b) Rotate the image shown below by backward mapping method with $\theta = 30^{\circ}$. (12)

1	5	2
6	3	4
7	9	5

   Use bilinear interpolation and allow the image to grow to fully contain the rotated image.
   c) Find the 4-point basis matrix of Discrete Fourier Transform (DFT) and then use it to find the DFT of [1 -1 1 -1]. (10)

8. a) Define image representation and description. Measure the entropy and uniformity of the following 3-bit image: (13)

1	2	0	7	5
6	2	3	4	3
4	0	2	1	5
4	7	6	2	4
5	6	4	3	3
5	4	1	2	4

   b) Explain the basic principle of Hough line detection mechanism. (09)
   c) What is optical flow? Explain Lucas-Kanade method of calculating optical flow. (13)